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Prof. Romain Rouvoy (University of Lille)
Dr. Quentin Perez (INSA Rennes)
Spirals & DiverSE project-teams

Subject: Application for PhD Position – Improving Sustainability in the Physical-to-Digital Twin Continuum (PC3)

Dear Prof. Rouvoy and Dr. Perez,

I am writing to express my strong interest in the PC3 PhD position focused on developing a sustainable middleware framework for adaptive digital twin deployment. My proven experience in optimizing complex machine learning architectures for low-power edge hardware makes me an ideal candidate to bridge the physical-to-digital twin continuum while rigorously minimizing environmental and computational footprints.

My commitment to sustainable AI engineering is best demonstrated by my work on the Industrial Digital Twin (LM2500 Gas Turbine). Rather than relying on cloud-heavy computations which incur massive energy costs, I specialized in Edge-Optimized AI. Specifically, I integrated highly efficient Mamba-SSM sequence models directly onto constrained edge hardware for real-time telemetry processing. By adapting the computation directly to the physical deployment constraints, I significantly reduced the energy requirements of continuous monitoring. This hands-on expertise directly translates to your goal of building an adaptive, eco-conscious middleware capable of dynamically scaling resources across the digital twin continuum based on the current context.

Furthermore, my background in Software Engineering provides a robust foundation for designing middleware architectures capable of seamlessly operating across heterogeneous hardware—from lightweight IoT sensors up to high-performance computing clusters. I am highly motivated to leverage these skills to build a deployment framework that autonomously optimizes resource usage, ensuring that the digital twin remains highly available and environmentally sustainable. I am excited about the opportunity to contribute to the Artemis platform. Please find attached my resume and academic transcripts. I welcome the opportunity to discuss my application further.

Sincerely,

Ismail Aissaoui